Z + b-tagged and Z + c-tagged jet: kinematics in Summer22

All POG corrections and SFs, plus BTV b- and c-tagging SFs on the leading jet

ParticleNet Loose (asked for), Medium and Tight, from one run

5 November 2026

What goes into these plots

Corrections and SFs (POG recipes)

Jets: JEC, JER, jet veto map, type-1 PUPPI MET. Leptons: muon and electron scale and smearing, ID, isolation and reco SFs, trigger SFs. Events: golden JSON, pileup weights, MET filters (new: they were missing before). The DY Z pT weights are now off.

working point	b-tag: PNet B >	c-tag: CvL >	and CvB >
Loose	0.047	0.054	0.181
Medium	0.245	0.160	0.306
Tight	0.673	0.492	0.259

Tagging is applied to the leading jet only

In MC each tagged jet gets the BTV SF for its true flavour. b-tag: particleNet_comb (b and c jets), particleNet_light. c-tag: particleNet_wc (c jets), particleNet_tnp (b jets), particleNet_light. b-tag and c-tag are separate selections, so a jet can pass both.

Yields in each region

Flavour shares are of all MC (DY + tt + VV). 'Tagged' is the fraction of Z + jet events that pass the tag.

region	data/MC Z→ee	data/MC Z→μμ	DY+b	DY+c	DY+light	tagged MC	tagged data
inclusive	0.95 ± 0.01	0.97 ± 0.01	4%	8%	87%	–	–
b-tag Loose	0.95 ± 0.03	0.95 ± 0.02	22%	21%	56%	17.9%	17.6%
c-tag Loose	0.93 ± 0.01	0.95 ± 0.01	1%	9%	90%	89.5%	88.2%
b-tag Medium	0.96 ± 0.05	0.94 ± 0.04	59%	17%	19%	6.0%	5.8%
c-tag Medium	0.94 ± 0.02	0.95 ± 0.02	3%	20%	77%	28.6%	28.2%
b-tag Tight	1.00 ± 0.07	0.90 ± 0.05	82%	5%	7%	3.5%	3.3%
c-tag Tight	0.92 ± 0.04	0.93 ± 0.03	10%	50%	39%	7.3%	7.0%

Flavour shares and tagged fractions are for Z→μμ; Z→ee is within a few percent.

The tagging SFs work: data and MC tag the same fraction of jets (within 1–2%).

Data/MC is about 0.95 in every region, the same as without tagging. The Loose c-tag keeps 89% of the events and they are 90% light, so it is not a charm selection. Medium (20% c) and Tight (50% c) are.

Tagging SFs and the overlap of the two tags

Mean SF of the tagged leading jet in DY MC, per true flavour. Overlap: tagged jets that also pass the other tag at the same working point.

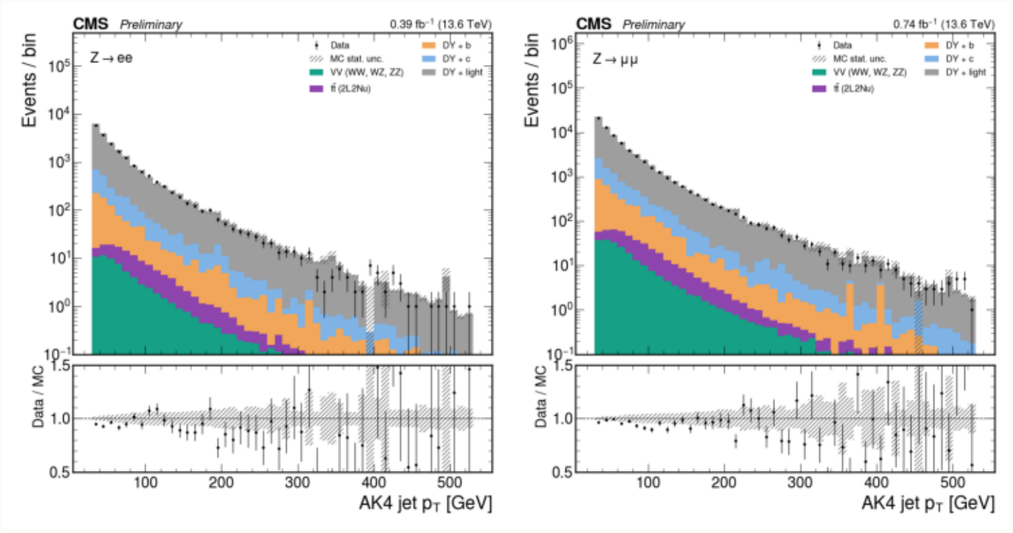
region	SF b jets	SF c jets	SF light jets	overlap, DY MC	overlap, data Z→μμ
b-tag Loose	0.95	0.94	1.21	77%	83%
b-tag Medium	0.94	0.93	1.29	33%	40%
b-tag Tight	0.92	0.92	1.27	4%	3%
c-tag Loose	0.78	1.00	1.06	13%	17%
c-tag Medium	0.77	0.97	1.09	7%	8%
c-tag Tight	0.79	0.83	1.16	2%	2%

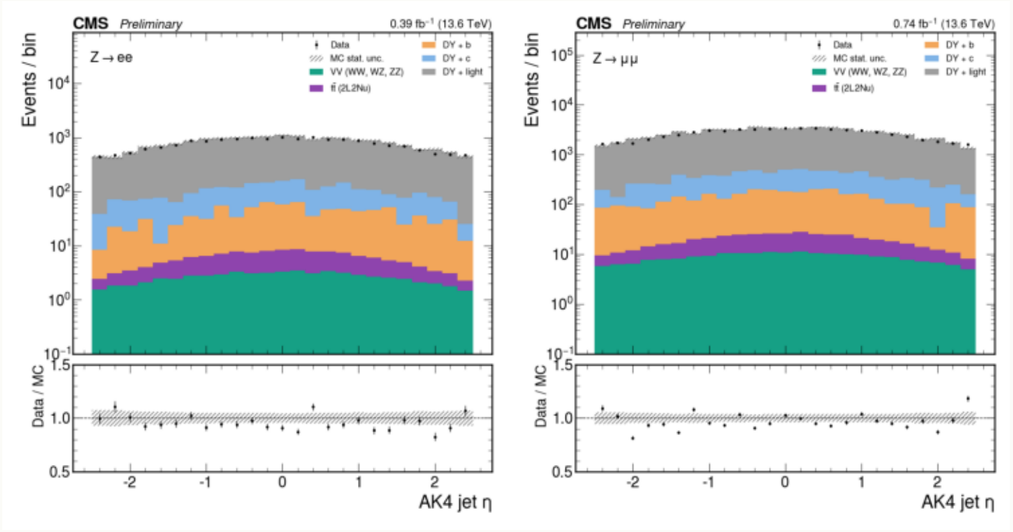
Light jets are mistagged more in data than in MC (SF 1.06–1.29).

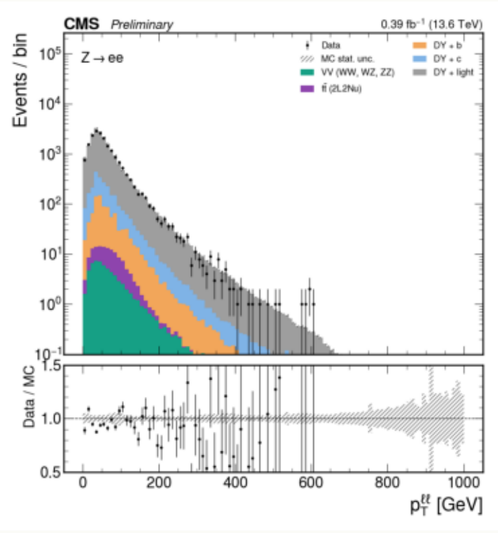
At Loose most b-tagged jets also pass the c-tag. At Tight the two selections barely overlap.

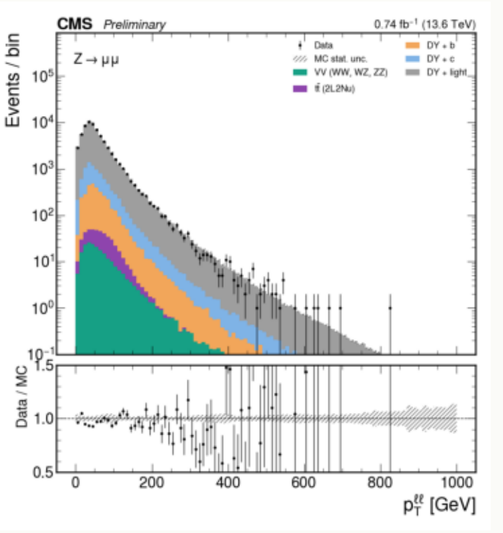
No tag requirement

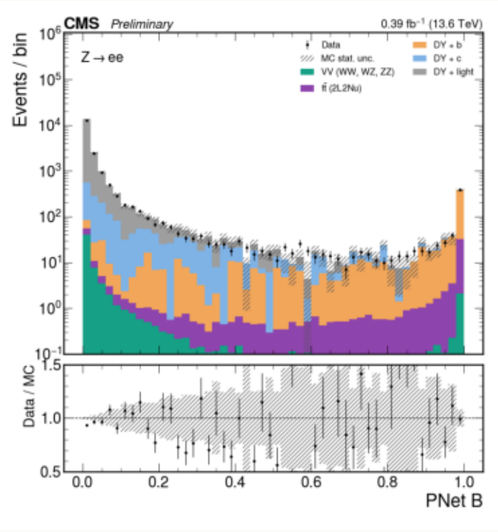
Z + at least 1 jet. Data/MC 0.95 (Z→ee) and 0.97 (Z→μμ).

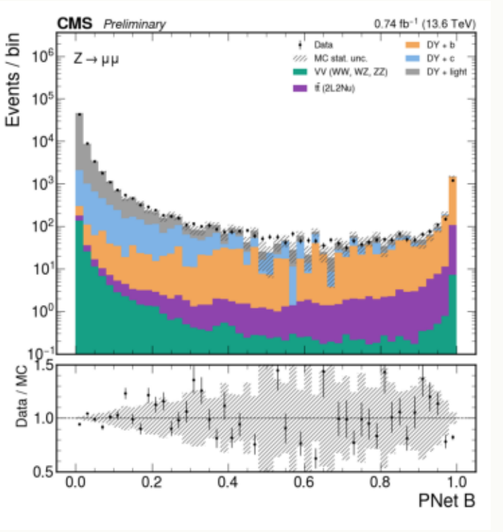






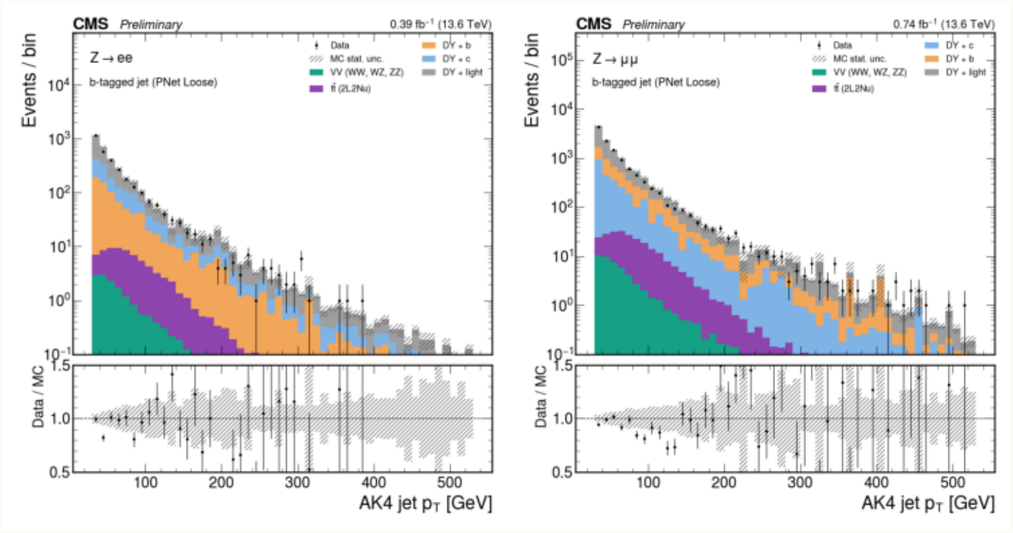


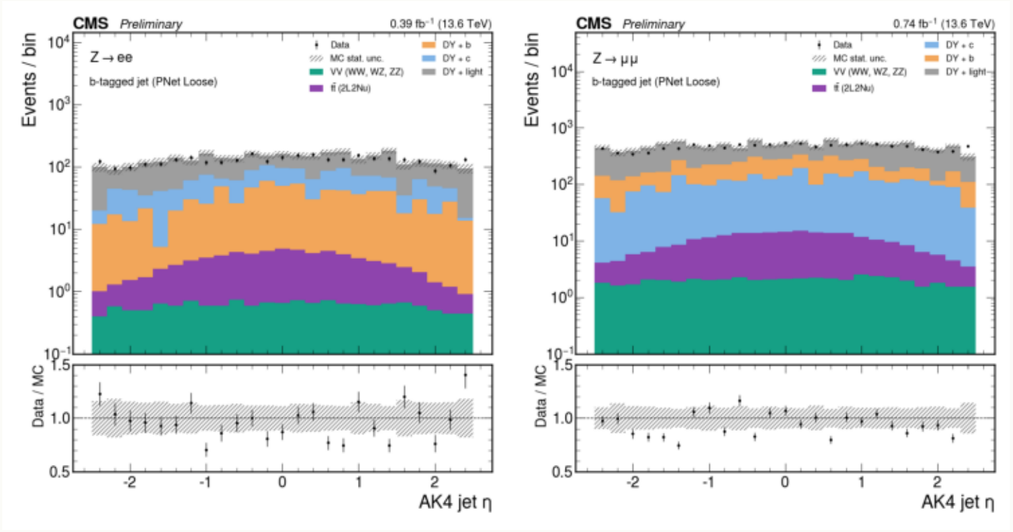


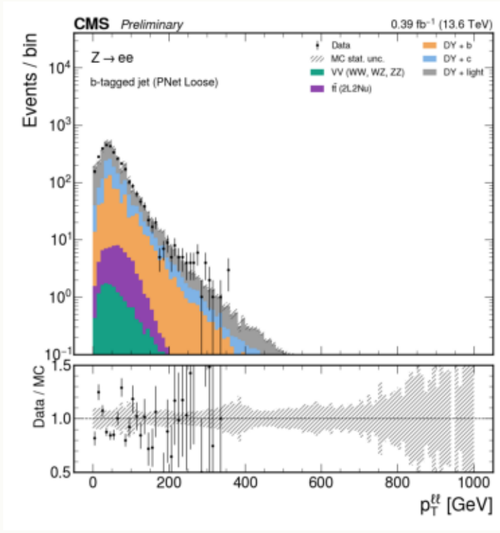


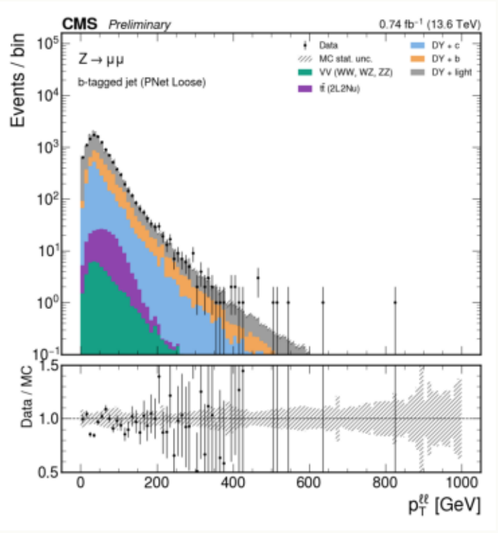
b-tag Loose

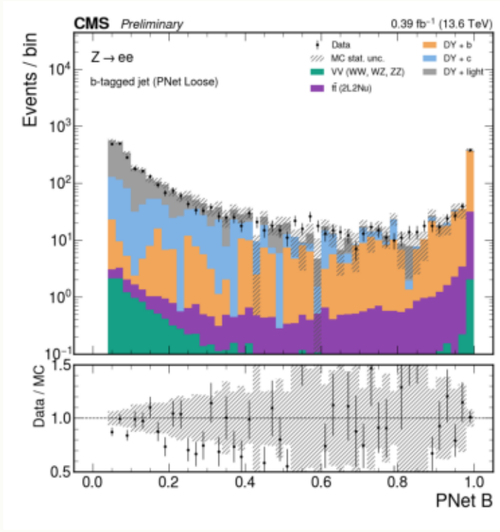
Leading jet PNet B > 0.047. 21–22% DY+b, 21–23% DY+c, 54–56% light. Data/MC 0.95.

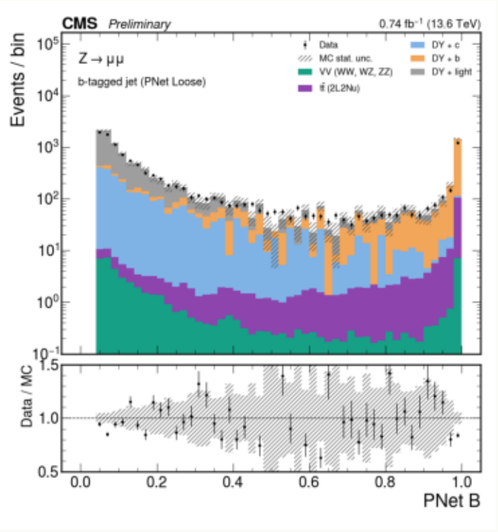






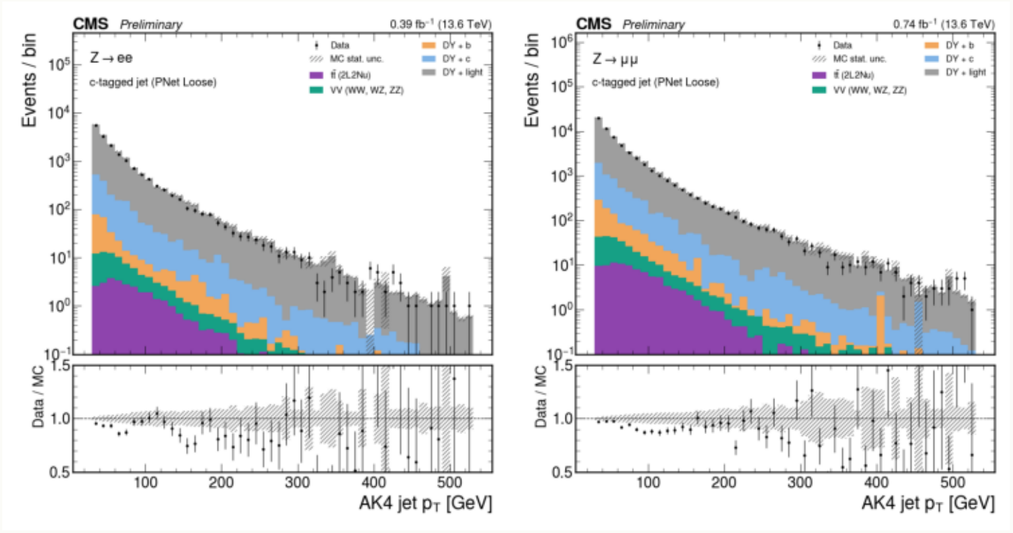


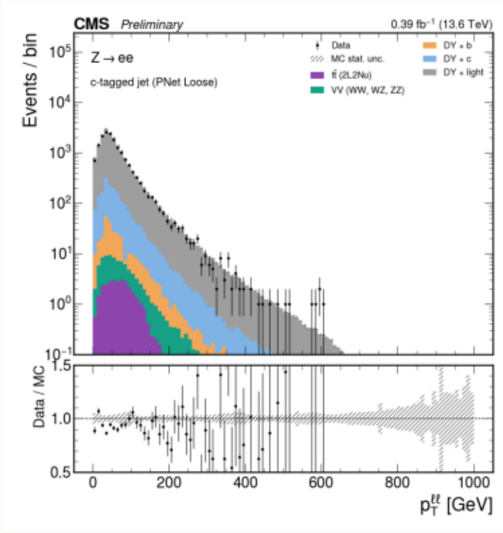


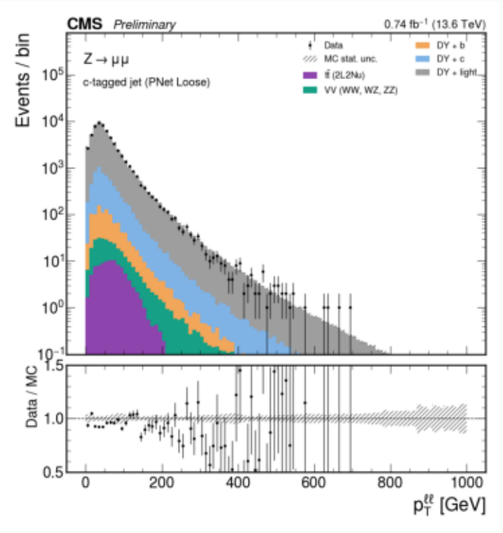


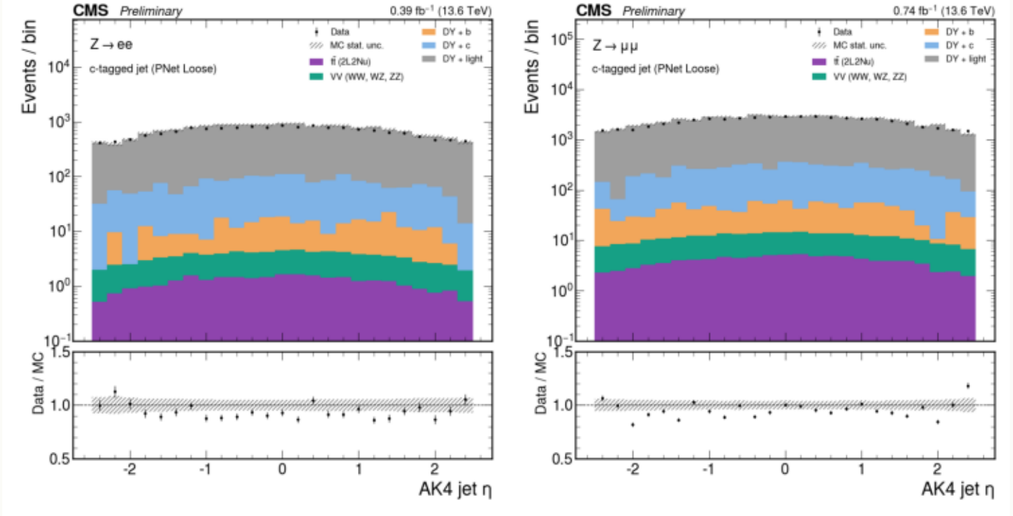
c-tag Loose

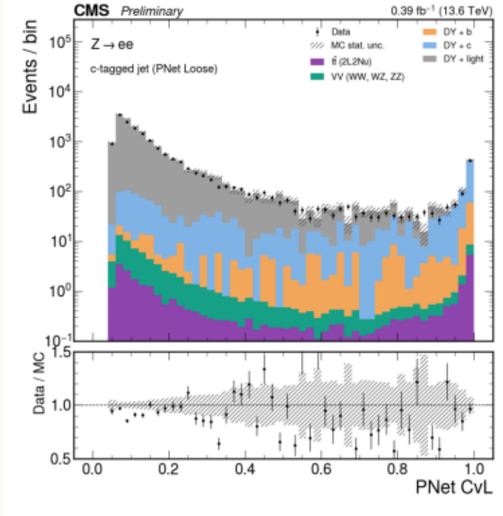
CvL > 0.054 and CvB > 0.181. 89% of the events pass; 9% DY+c, 90% light. Data/MC 0.93–0.95.

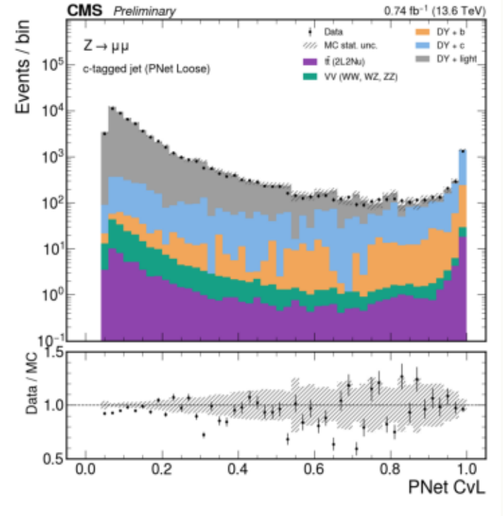






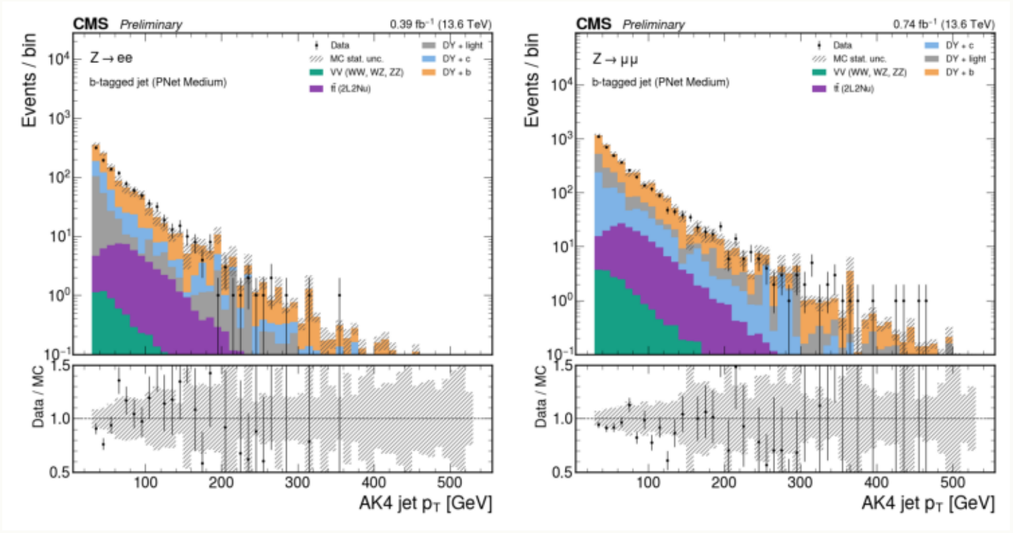


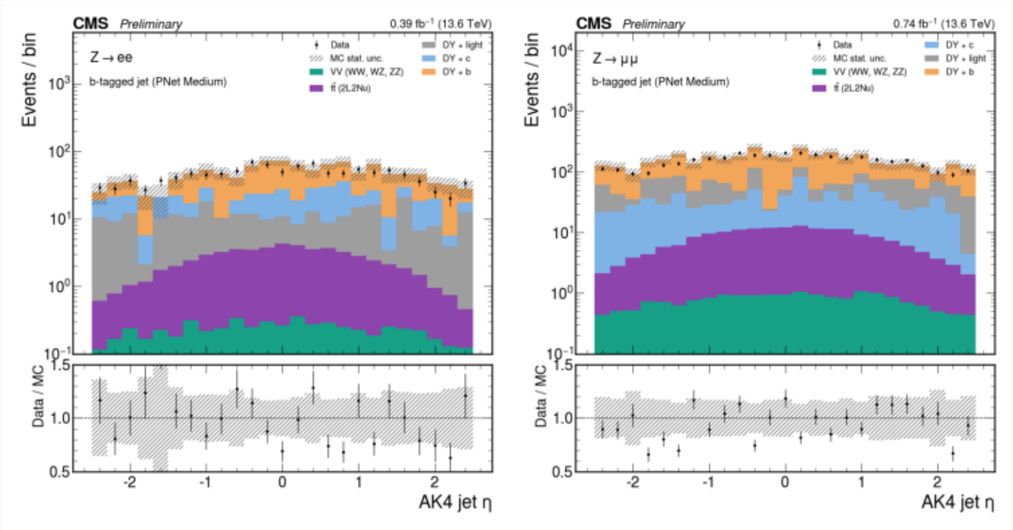


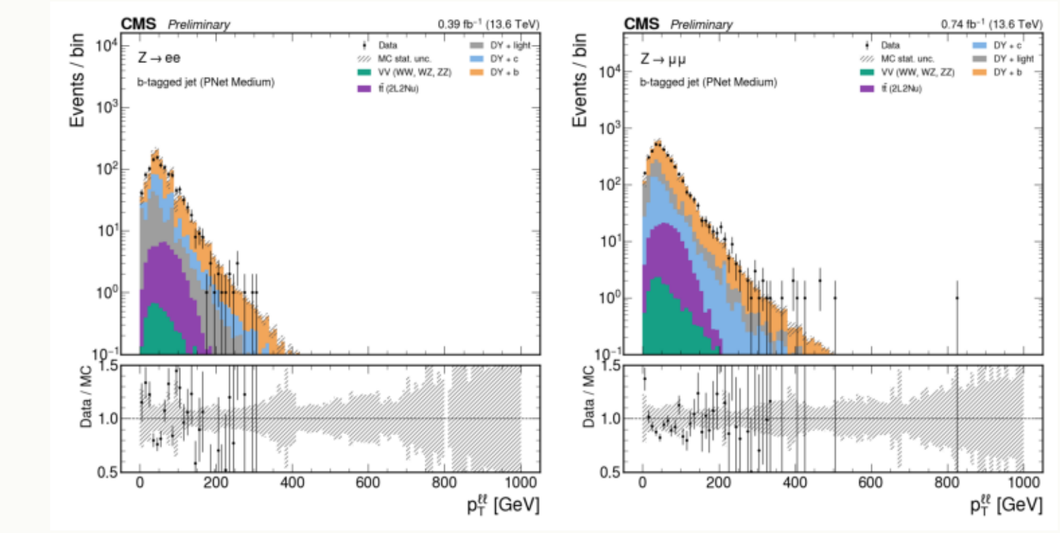


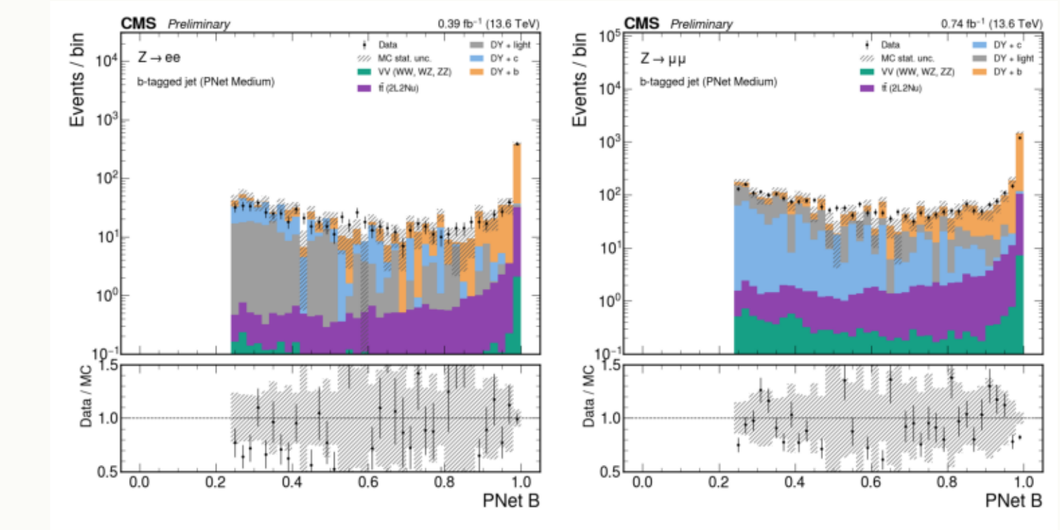
b-tag Medium

PNet B > 0.245. 55–59% DY+b, about 4% tt. Data/MC 0.94–0.96.



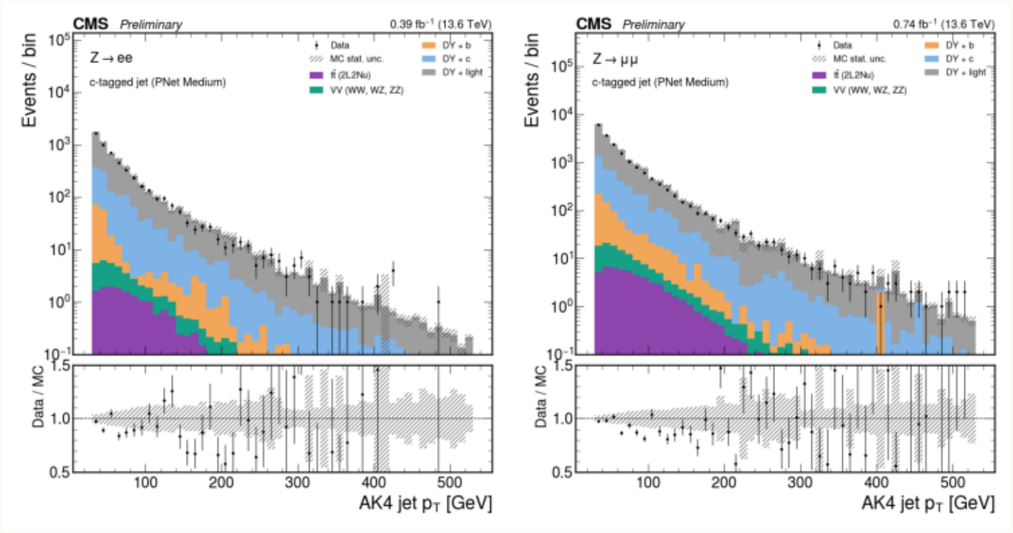


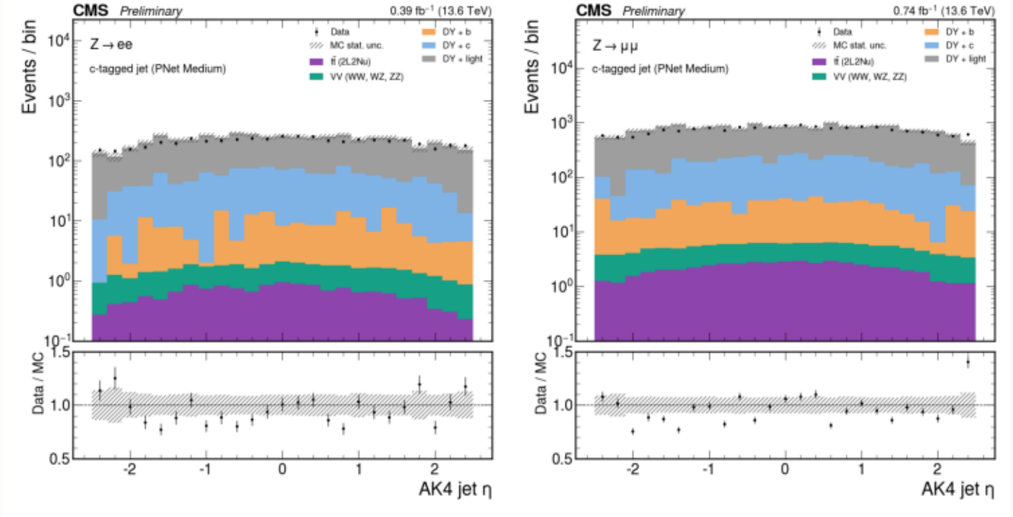


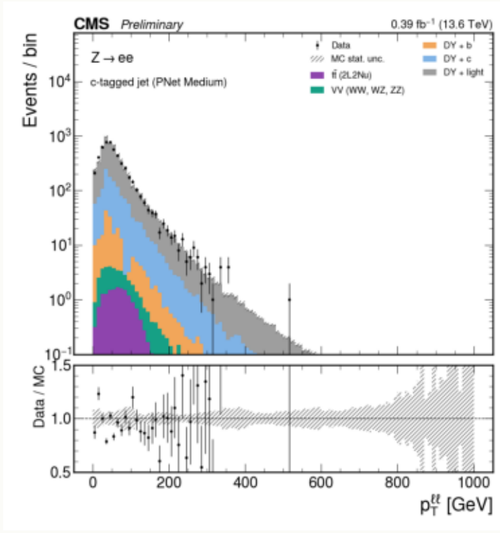


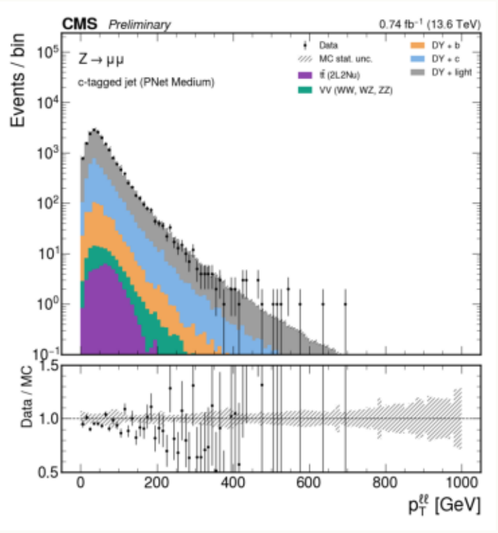
c-tag Medium

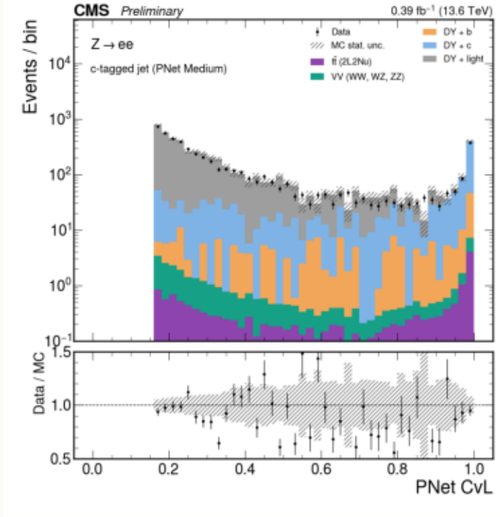
CvL > 0.160 and CvB > 0.306. 20% DY+c, 77% light. Data/MC 0.94–0.95.

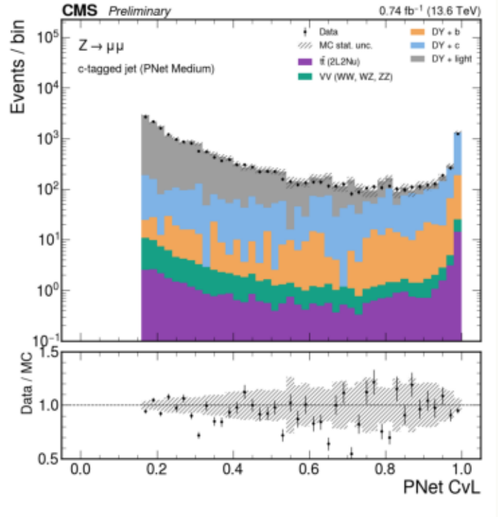






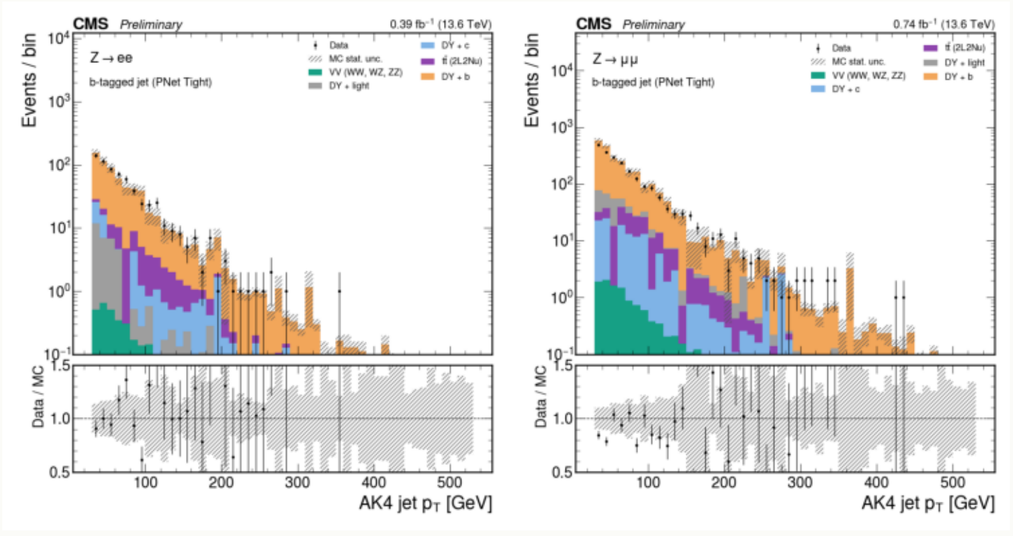


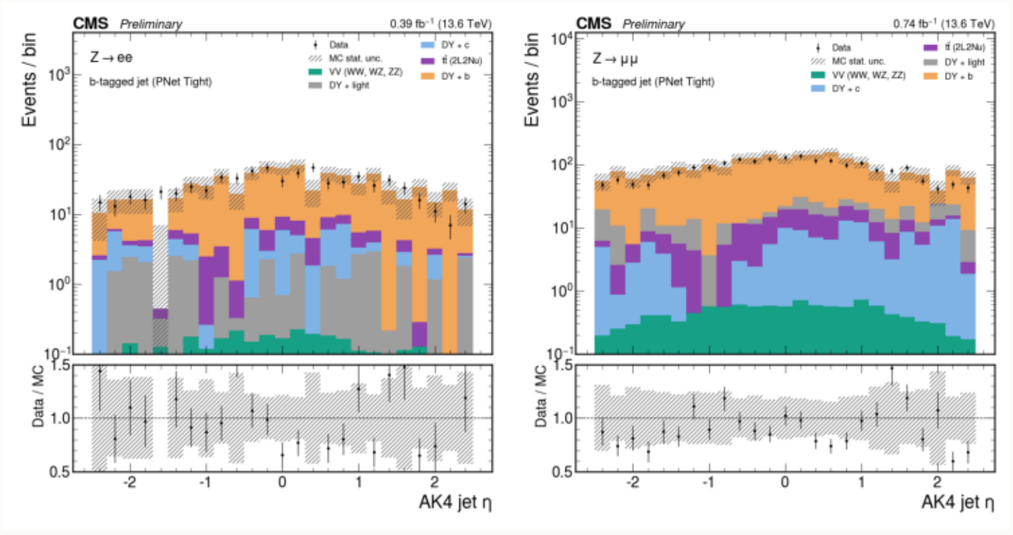


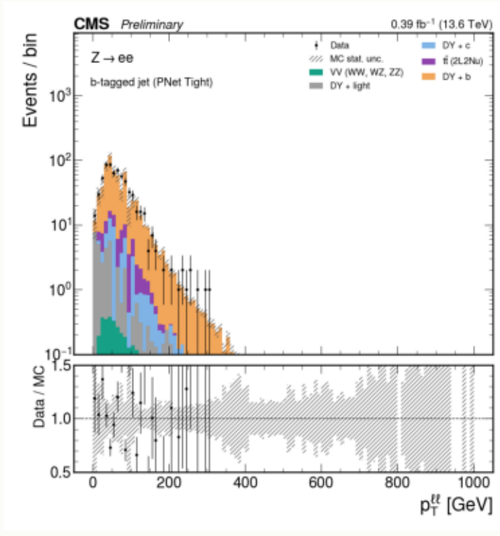


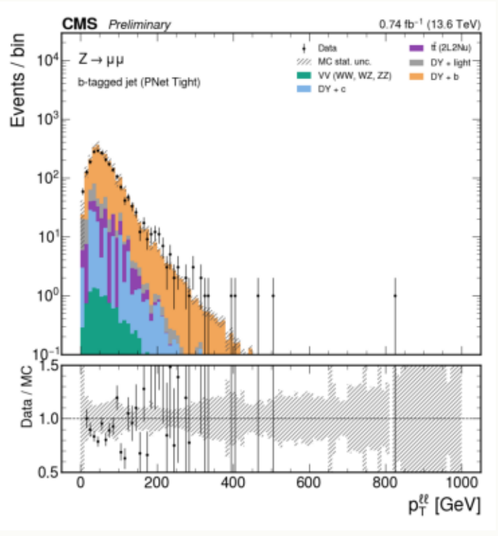
b-tag Tight

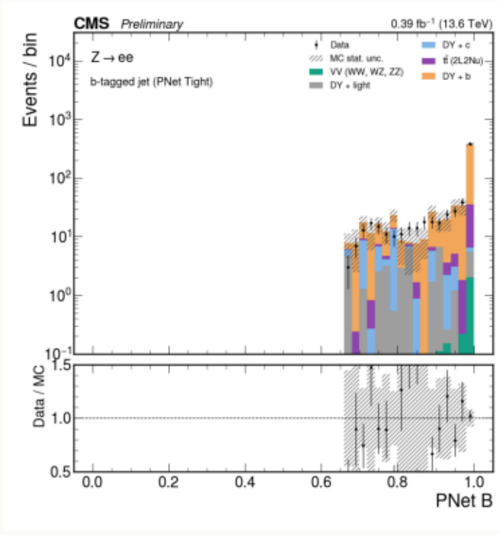
PNet B > 0.673. 82% DY+b, 6–7% tt. Data/MC 1.00 ± 0.07 (Z→ee), 0.90 ± 0.05 (Z→μμ).

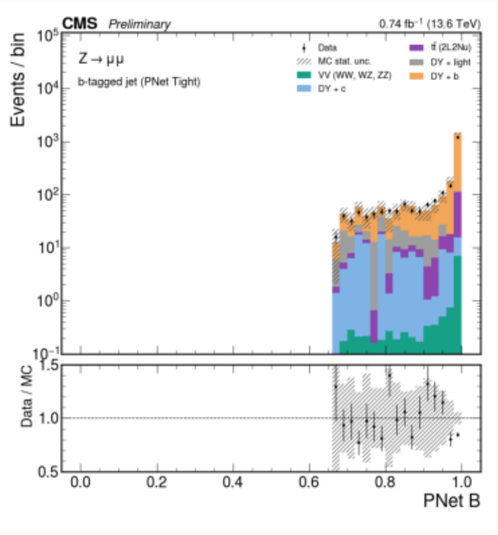






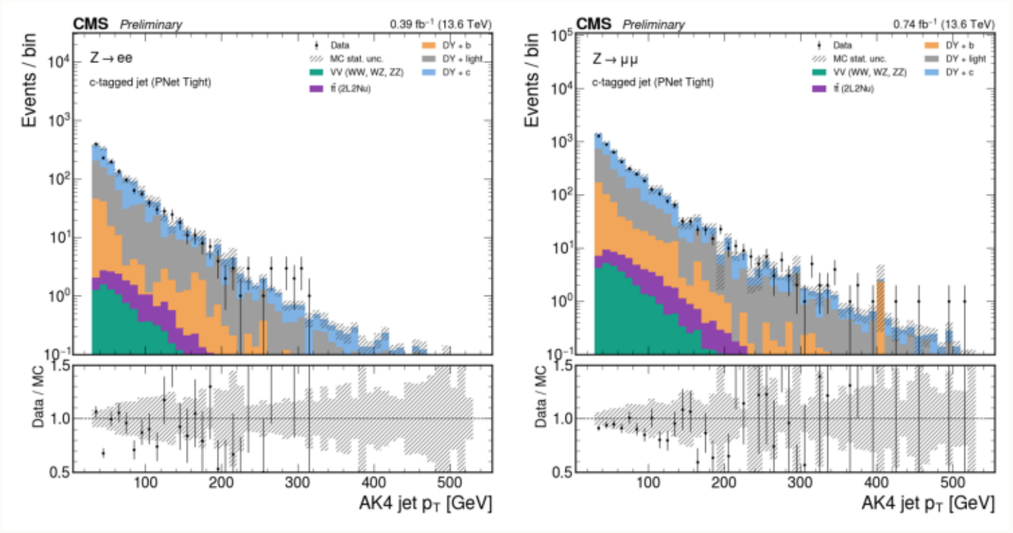


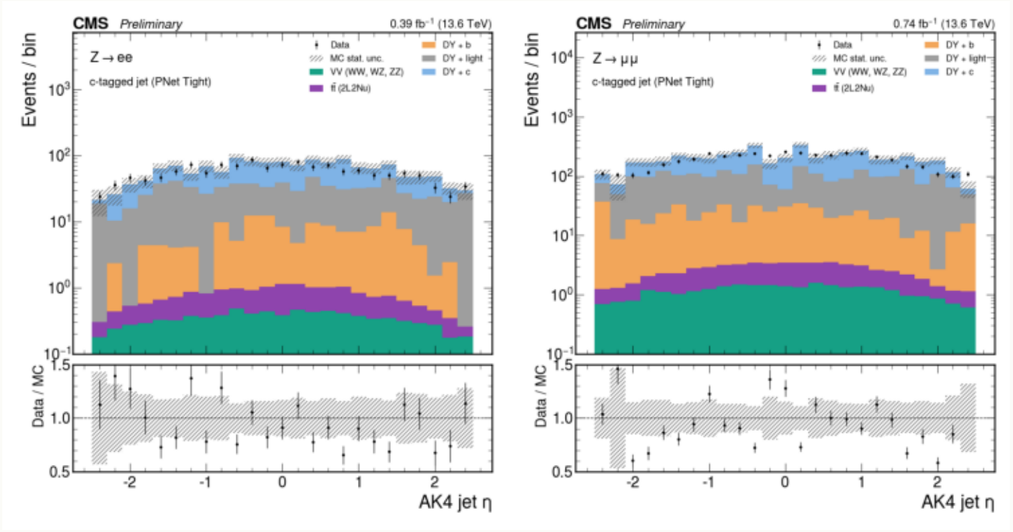


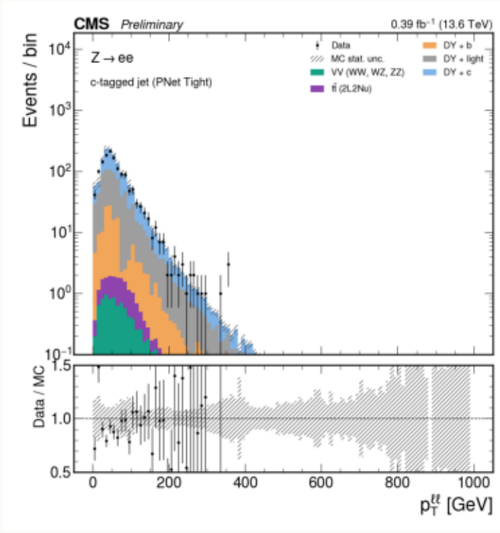


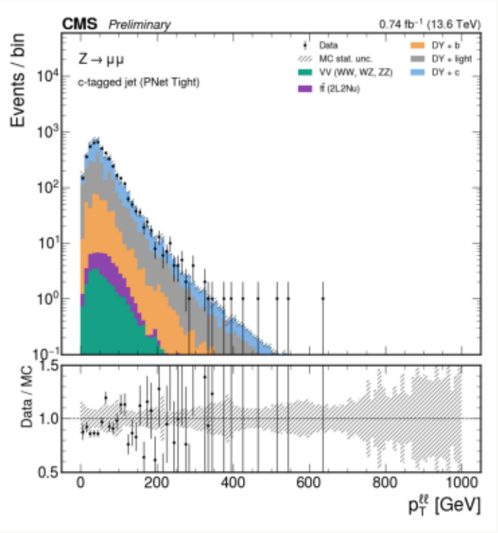
c-tag Tight

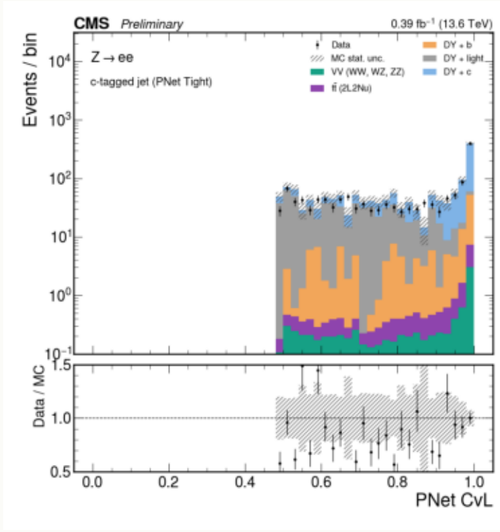
CvL > 0.492 and CvB > 0.259. 50% DY+c, 39% light, 10% DY+b. Data/MC 0.92–0.93.

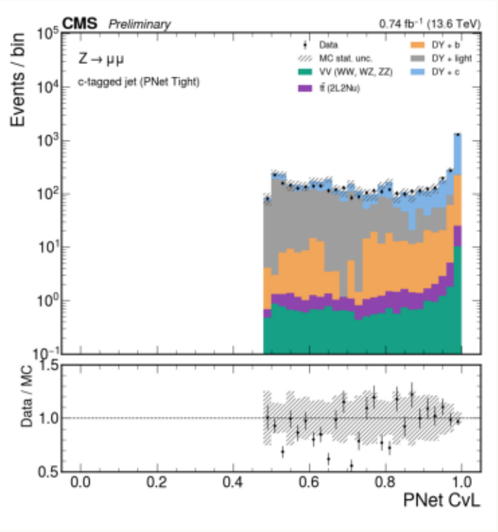












Summary

All kinematic plots now have every POG correction, plus b- and c-tagging SFs.

62 plots per region, in hists_zbzc_run3_tag/Summer22/plots_<region>/. MET filters added; DY Z pT weights off.

Data and MC tag the same fraction of jets, within 1–2%, at every working point.

Data/MC is about 0.95 in all regions, as without tagging. Errors here are statistical only.

Loose c-tag is almost no selection (89% of events, 90% light).

For a charm-enriched sample, Medium (20% c) or Tight (50% c) is better. b-tag Medium gives 55–59% b, Tight 82%.

Next

Add the SF up/down uncertainties as a band. Same plots for 2022 E–G and 2023.